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Fine-Tuning the Prithvi Foundation Model for Crop-Type and Health Mapping in Smallholder Farms

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Abstract

Artificial intelligence is rapidly reshaping how we monitor and manage agriculture from space. In this study, we fine-tune and evaluate a state-of-the-art vision foundation model, Prithvi-EO-2.0, for crop type classification and crop health monitoring across smallholder farms in Telangana, India. Leveraging temporally stacked satellite imagery, we assess the effectiveness of different fine-tuning strategies and address challenges related to fragmented landholdings and limited ground truth data. Our results demonstrate that fully fine-tuning the model significantly improves crop classification accuracy but requires careful handling of class imbalance to prevent overfitting. We discuss limitations arising from spatial resolution constraints and propose future directions using higher-resolution alternatives, emphasizing the potential of adapting large-scale pretrained models to deliver scalable and practical precision agriculture solutions in data-limited, heterogeneous smallholder systems.

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1. Introduction

The integration of Artificial Intelligence (AI) into Earth Observation (EO) systems has significantly advanced agricultural monitoring, enabling enhanced precision in crop health assessment, yield prediction, and resource man-

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¹ Code and dataset available at <https://gitlab.hhi.fraunhofer.de/dawoud/finetune-prithvi-for-smallholders.git>. The dataset is licensed under CC-BY SA 4.0.

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agement. Machine learning (ML) and deep learning (DL) methodologies now facilitate the analysis of vast geospatial datasets, supporting data-driven decisions tailored to local agricultural conditions [9]. This shift toward precision agriculture aims to maximize yields while minimizing environmental impacts, addressing critical challenges such as climate change, resource scarcity, and global food security [16, 22].

Recent advancements in AI, particularly the emergence of transformer-based architectures, have further impacted agricultural applications, improving tasks such as crop classification, yield estimation, and disease detection [19]. In parallel, the introduction of vision foundation models, large models pre-trained on diverse EO datasets, has significantly enhanced the ability to analyze complex spatio-temporal patterns in remote sensing data [7, 12]. These models, characterized by large-scale self-supervised pre-training and high adaptability, offer robust, generalizable representations across diverse EO tasks. However, their real-world effectiveness, especially in smallholder agricultural systems characterized by fragmented land holdings and heterogeneous practices, remains underexplored.

In India, smallholder and marginal farmers, those who own less than two hectares of land, represent approximately 86.2% of all farmers, according to the 2015 agriculture census [3]. This significant demographic highlights the urgent need for scalable, adaptable AI-driven EO solutions tailored to the unique challenges faced by smallholder farmers.

To address this gap, this study investigates the application of the vision foundation model Prithvi-EO-2.0 [17] for crop type classification and crop health assessment among small farms in Telangana, India. Specifically, we fine-tune the model on a newly curated dataset containing detailed annotations of farm boundaries, crop types, management practices, and health labels, and systematically evaluate two fine-tuning strategies: using the pretrained encoder as a frozen feature extractor versus fully fine-tuning all model parameters.

Our experiments leverage temporally stacked satellite imagery to capture crop phenological stages critical for accurate classification. We also address class imbalance challenges and analyze model performance relative to field size, highlighting limitations stemming from the spatial resolution of available satellite data.

Overall, this work contributes:

- A systematic evaluation of fine-tuning strategies for adapting the Prithvi-EO-2.0 vision foundation model to crop type and crop health classification tasks in smallholder agricultural settings.
- An assessment of using temporally stacked HLS imagery to model crop phenology with limited ground truth data.
- A discussion of limitations arising from satellite resolution constraints when monitoring fragmented, small-scale farms, with proposed future directions involving higher-resolution Sentinel-2 alternatives.

By aligning model development closely with the practical realities of smallholder agriculture, this study advances the understanding of how vision foundation models can support scalable, farmer-centered precision agriculture solutions.

2. Related Work

Remote sensing has become an indispensable tool for monitoring agricultural systems, enabling large-scale crop yield forecasting, classification, and health assessment. The last decade has witnessed rapid advances in leveraging temporal satellite data with deep learning techniques. This section provides an overview of related work, structured chronologically and thematically, highlighting key trends from early neural models to recent foundation models.

2.1. Early Deep Learning for Temporal Crop Monitoring

Early deep learning approaches primarily explored how temporal signals in satellite imagery could be captured using convolutional and recurrent neural networks. You *et al.* [21] pioneered a hybrid CNN-LSTM architecture augmented with a Gaussian Process for crop yield prediction, utilizing dimensionally-reduced histograms from multi-spectral time series. Their method significantly outperformed earlier models relying on hand-crafted features.

Rußwurm and Körner [15] extended this direction with an encoder-decoder model using convolutional LSTM layers to learn phenological patterns from Sentinel-2 data, achieving high accuracy in crop classification. These works emphasized the importance of explicitly modeling temporal dynamics, an advantage over static 2D CNNs applied to

composite imagery. Later, Zhong *et al.* [23] demonstrated that 1D temporal CNNs, which convolve across the time axis, can rival or surpass RNNs in multi-temporal crop classification.

2.2. Semantic Segmentation for Crop Type and Health Mapping

Beyond classification, semantic segmentation techniques have been applied to delineate within-field variability, such as stressed versus healthy regions. The introduction of fully convolutional networks (FCNs) and U-Net architectures marked a turning point in high-resolution agricultural segmentation tasks.

Chiu *et al.* [4] released the Agriculture-Vision 2020 dataset, a large-scale benchmark for detecting common anomalies such as nutrient deficiencies, water stress, and weed infestation in aerial imagery. Using Feature Pyramid Networks (FPNs), they performed pixel-wise segmentation while addressing domain-specific challenges like sparse labels and large image sizes.

Rustowicz *et al.* [13] proposed a temporal semantic segmentation approach for African croplands, comparing a 2D U-Net combined with ConvLSTM to a fully 3D U-Net that treats time as an additional spatial dimension. Their findings showed that incorporating temporal sequences from Sentinel-1 and Sentinel-2 significantly improved segmentation accuracy, with ConvLSTM-based U-Nets outperforming 3D U-Nets in some scenarios.

Similarly, Adrian *et al.* [1] demonstrated that a 3D U-Net trained on fused radar and optical satellite time series enables more accurate within-field crop mapping. These studies reflect a broader shift from static, single-date image analysis to leveraging multi-temporal stacks for more robust and dynamic segmentation.

2.3. Transformers and Vision Foundation Models in Remote Sensing

More recently, transformer-based architectures and large pre-trained “foundation models” have emerged as dominant paradigms in remote sensing. Originally developed for natural language processing, transformers are now widely used in Earth observation due to their capacity to model long-range dependencies through self-attention.

The Vision Transformer (ViT) [6] laid the groundwork for this shift. Wang *et al.* [18] introduced Cropformer, a transformer-based model for multi-scenario crop classification, utilizing self-supervised pretraining on unlabeled multi-temporal data, followed by fine-tuning on specific crop labels. This approach captured long-term temporal dependencies with fewer annotated samples.

Multi-modal models have also gained traction, fusing satellite imagery with auxiliary data like weather and soil. Lin *et al.* [10] proposed MMST-ViT, a climate-aware vision transformer that jointly ingests satellite time series and meteorological data for more accurate yield prediction under varying climate scenarios. Their model captures complex interactions between environmental stressors (e.g., heat, drought) and crop phenology.

Jakubik *et al.* [7] introduced Prithvi, a 100M-parameter transformer pre-trained on 1 TB of Harmonized Landsat-Sentinel-2 (HLS) imagery via masked image modeling. This foundation model treats multi-temporal satellite data as video-like sequences, enabling efficient fine-tuning for diverse downstream tasks such as flood detection, burn scars, and multi-crop segmentation, often matching or exceeding task-specific models with far less labeled data.

More recently, Xiong *et al.* [20] introduced DOFA, a domain-optimized foundation model that achieves state-of-the-art performance across a range of Earth observation tasks. DOFA leverages extensive pretraining and task-specific architectural innovations; however, its primary focus remains on spatial representations, with less emphasis on capturing temporal dynamics compared to models like Prithvi-2.

In this work, we adapt the Prithvi-EO-2.0 foundation model [17] for crop type mapping and health assessment in smallholder farms across Telangana, India. While recent evaluation [17] suggests that DOFA marginally outperforms Prithvi-2 in certain tasks, we opt for Prithvi-2 due to its explicit temporal modeling capabilities, which are critical for our multi-temporal agricultural monitoring tasks. Our study demonstrates the effectiveness of foundation models in supporting precision agriculture, especially in data-sparse regions, and highlights their potential to deliver scalable, general-purpose solutions across agricultural landscapes.

3. Data Preparation

The dataset used in this work is based on the tabular data provided by [5] and is hosted on the Agriculture Data Exchange (ADeX) platform. Originally developed for the Telangana Crop Health Challenge [24], it focuses on crop

health monitoring in the Indian state of Telangana (see Fig. 1). Serving as the ground truth, the dataset includes detailed geographic farm boundaries and comprehensive metadata such as crop types, cultivation practices, irrigation methods, crop rotation patterns, and sub-district-level agricultural statistics. In addition, it provides crop-specific attributes, such as planting and harvest dates, irrigation frequency, and yield targets, which are valuable for agricultural modeling.

The original tabular dataset contains over 10,608 farm entries. However, after applying our selection criteria to create the satellite-based training samples, only 601 farm parcels remained, with observation dates ranging from February 2023 to August 2024. This reduction is due to strict requirements for metadata completeness, spatial boundary accuracy, and satellite data availability.

A critical step in the data preparation pipeline involves generating labeled training samples (chips) by integrating this reference dataset with satellite imagery. Specifically, the process includes: 1) Creating raster segmentation masks based on the crop labels in the ground truth dataset. 2) Aligning and merging these masks with satellite imagery to generate training-ready image chips.

The labeled dataset encompasses eight crop-type classes: ‘Paddy’, ‘Cotton’, ‘Bengal Gram’, ‘Groundnut’, ‘Chillies’, ‘Red Gram’, ‘Maize’, and ‘Background’. Supplementary crop health labels: ‘Healthy’, ‘Stressed’, ‘Diseased’, and ‘Pests’ are also provided in the metadata.

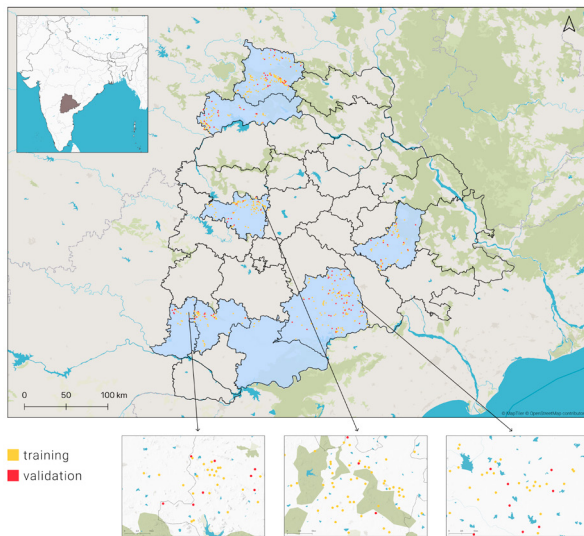


Fig. 1. Location of agricultural parcels in Telangana, India. Yellow dots represent training set parcels, and red dots indicate validation set parcels.

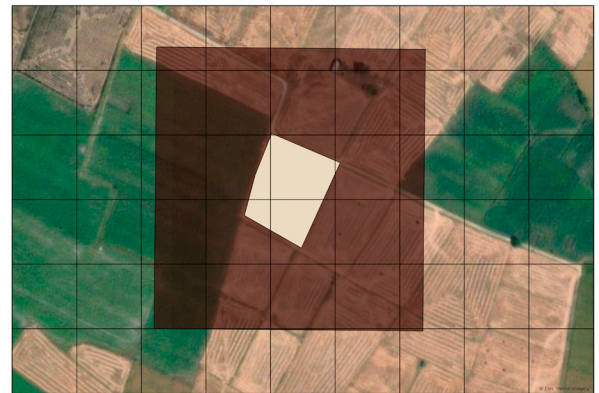


Fig. 2. Example of a training chip with a 50 m grid overlay. The beige central area represents the farm boundary, and the surrounding dark-shaded region denotes the buffer zone labeled as ‘Background’.

A buffer zone was added around each farm boundary to meet the model’s input requirements and provide additional spatial context (see Fig. 2). Pixels within the boundary, which are shown in beige, were labeled based on crop type or health. In contrast, the pixels in the surrounding buffer area, represented in brown, were assigned the ‘Background’ class. These background pixels were ignored during the training and validation processes because ground truth data for this surrounding area is not available.

3.1. HLS Data Description

For the satellite imagery used in this study, we selected the Harmonized Landsat Sentinel-2 (HLS) dataset, which integrates observations from NASA/USGS Landsat 8 and 9 missions as well as ESA’s Sentinel-2A and 2B missions. This dataset was also employed during both the pre-training and fine-tuning stages of the Prithvi-EO-2.0 model, ensuring continuity between training phases.

To ensure high-quality data preparation, the processing pipeline included several steps. We downloaded the HLS imagery from the Google Earth Engine public data archive and clipped each image to the spatial boundaries defined in the ground truth dataset. Temporal ranges were set based on the sowing and harvesting dates specified in the accom-

panying metadata. For further quality assurance, images were filtered based on two main criteria: pixel completeness, requiring at least 97% of pixels to be present, and cloud cover, allowing no more than 10%.

For each farm location, we selected three representative time points: early, mid-, and late season to capture key phenological stages of crop development. To standardize inputs and ensure sufficient spatial coverage within each training chip, we resampled the imagery to a fixed size of 446×446 pixels at 0.5 meters per pixel. This high-resolution upsampling was necessary because, at the original 30-meter resolution of HLS imagery, many smallholder farm plots would occupy only a few pixels, often too few for the model to learn meaningful spatial patterns. By resampling to 0.5 meters, we expanded each chip's spatial footprint, providing more pixels per farm boundary and better alignment with labeled plot geometries, which improved the model's ability to learn fine-grained distinctions.

We used nearest-neighbor interpolation for resampling due to its simplicity and efficiency. Although pan-sharpening could theoretically offer improved spectral and spatial fidelity on other satellite data, it is not practical for HLS data. This is because HLS already harmonizes data from Sentinel-2 and Landsat missions by resampling all bands to a common 30-meter resolution, effectively discarding the native higher-resolution (10–20 meter) bands of Sentinel-2. As a result, pan-sharpening cannot recover spatial detail that has already been lost during the harmonization process, making it an unsuitable enhancement technique for HLS-based workflows.

Although the HLS dataset provides 15 spectral bands, we retained only the six bands common to the Sentinel-2 and Landsat missions to maintain compatibility with the model's pretraining configuration. These bands included Blue, Green, Red, Near-Infrared (NIR), Shortwave Infrared 1 (SWIR1), and Shortwave Infrared 2 (SWIR2). The selected channels were stacked across the three time points, forming a three-frame composite with a final shape of (446, 446, 18), where the dimensions represent spatial extent and six channels per time step across three times. If a farm has more than three images, we select the earliest image, the middle image, and the last image from the time between the sowing and harvesting dates. This method ensures that the three images encompass the entire crop lifecycle.

Each training chip corresponds to a unique, non-overlapping spatial instance extracted from the reference dataset, serving as an independent input sample for the model.

4. Ablation Study and Results

This section presents results from fine-tuning the vision foundation model **Prithvi-EO-2.0-300M** for crop type and crop health classification tasks in Telangana. Since the original model lacks domain-specific agricultural training, we treat it as a pretrained encoder and systematically evaluate different fine-tuning strategies.

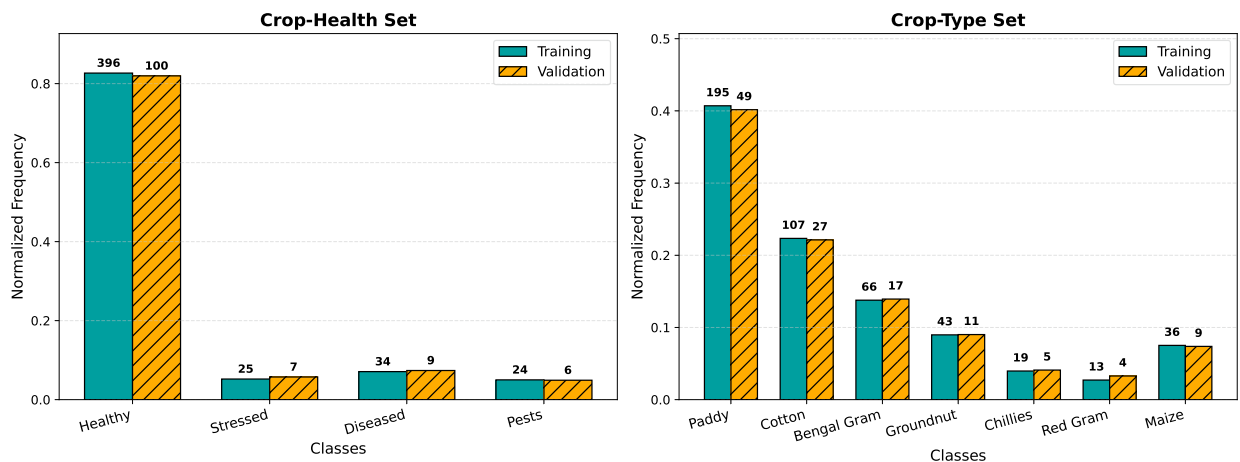


Fig. 3. Class distribution across the training and validation datasets shows the overall percentage of each class and the number of actual samples in both these sets. Left: crop-health set. Right: crop-type set.

4.1. Fine-tuning Strategies

We conducted an ablation study comparing two primary fine-tuning strategies: (1) **Frozen Encoder**, in which only the decoder and classification head were trained while encoder weights remained fixed, and (2) **Full Fine-tuning**, in which all layers (encoder and decoder) were updated. Additionally, we trained a reference model from scratch using the same architecture for comparison.

Evaluation was conducted on two distinct tasks: **Crop Type Classification**, predicting one of eight crop types per input sample, and **Crop Health Classification**, identifying crop conditions categorized as *Healthy*, *Stressed*, *Diseased*, or *Pests*.

All models utilized consistent dataset splits for comparability, with class distributions illustrated in Figure 3. To address class imbalance, experiments compared Cross-Entropy, Focal Loss [11], and Dice Loss [14]. Training was optimized with AdamW, a learning rate of 0.0001 for 100 epochs.

Input data comprised three temporal slices from HLS imagery, representing early, mid, and late-season periods. Data augmentation methods, including random flipping and warp-affine translation, were applied to minimize spatial biases.

4.2. Class-Imbalance Analysis

Class imbalance posed significant challenges across both classification tasks. In crop type classification, a pronounced skew was evident: Paddy and Cotton comprised over 60% of samples, while minority crops like Red Gram and Chillies each accounted for less than 5%. For crop health classification, the *Healthy* category dominated ($\geq 80\%$), with other conditions under-represented ($< 10\%$).

To mitigate these imbalances, we employed Focal and Dice Loss functions to prioritize minority classes. Additionally, class-balanced weighted random sampling ensured more equitable representation during mini-batch formation.

4.3. Results Summary

Performance evaluation employed macro F1 Score, Intersection over Union (IoU), and Accuracy metrics. Table 1 summarizes results using Dice Loss with the full dataset. Full fine-tuning consistently outperformed the frozen encoder and models trained from scratch across tasks. Notably, for crop type classification, full fine-tuning achieved a superior F1 score of 0.79. For crop health classification, improvements were marginally better than the model trained from scratch (F1 score 0.32 vs. 0.31). Learning curves (Figure 4) reinforced these findings, showing that the fully fine-tuned model converged faster and generalized better in crop type classification, whereas accuracy in crop health classification masked underlying overfitting on the majority class.

We further explored the benefits of pretraining under constrained data scenarios, detailed in Figure 6. Pretrained models demonstrated substantial advantages over models trained from scratch at lower dataset fractions (20%–40%) for crop type classification, clearly highlighting their utility in data-limited contexts. For crop health classification, pretrained models notably outperformed training from scratch within intermediate dataset fractions (40%–60%), underscoring their effectiveness in moderately limited data conditions. However, performance improvements plateaued beyond 60% dataset utilization in crop health classification, whereas models trained from scratch continued to benefit from increasing data up to 80%, suggesting diminishing returns for pretrained models when ample data is available.

Table 2 illustrates the comparative efficacy of various loss functions, confirming Dice Loss as the most effective, particularly for enhancing recall of minority classes.

Our findings highlight: (1) the critical advantage of full fine-tuning, particularly for balanced tasks, while emphasizing careful management for imbalanced scenarios; (2) robust performance gains from pretrained models in scenarios with limited training data; and (3) Dice Loss's effectiveness in addressing class imbalance, significantly benefiting the crop health classification task.

Representative inference examples (Figure 7) visually confirm the model's practical ability to accurately delineate crop boundaries and assess crop health conditions.

Table 1. Comparison of fine-tuning strategies for crop type and crop health classification using Dice Loss. The performance metrics (Macro F1 Score, IoU, and Accuracy) are based on the best model evaluated over the validation dataset.

Task	Strategy	F1 Score	IoU (Jaccard)	Accuracy
Crop Type Classification	From Scratch	0.63	0.49	0.69
	Frozen Encoder	0.56	0.50	0.68
	Full Fine-tuning	0.79	0.67	0.82
Crop Health Classification	From Scratch	0.31	0.22	0.68
	Frozen Encoder	0.20	0.17	0.52
	Full Fine-tuning	0.32	0.25	0.71

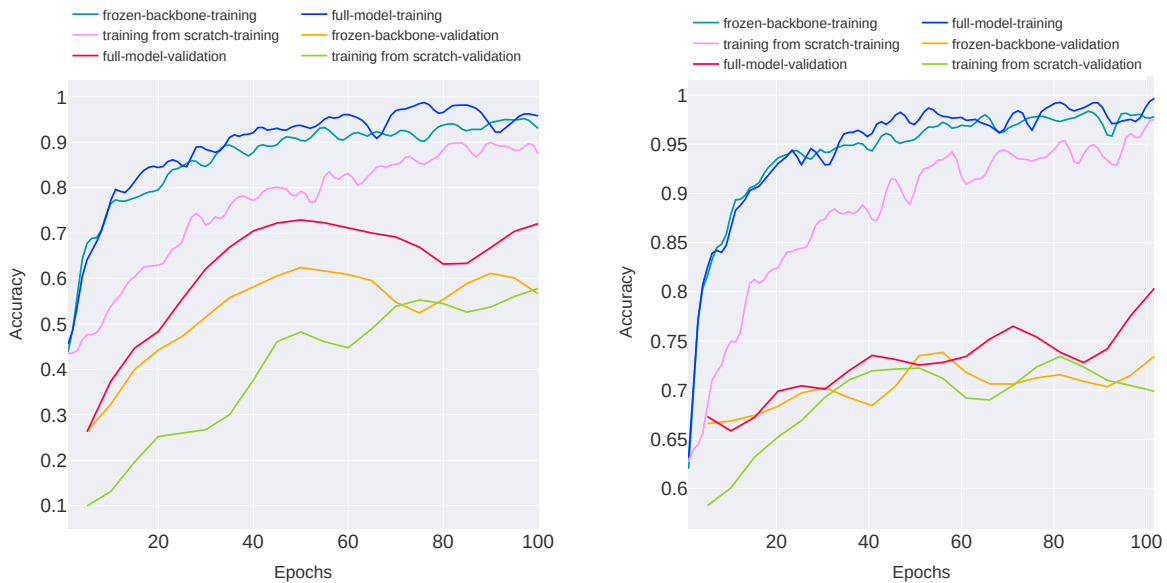


Fig. 4. Multi-class accuracy across training epochs for crop health (left) and crop type (right) classification tasks. Although the fully fine-tuned model appears superior in accuracy for crop health, results in Table 1 indicate overfitting. In contrast, for crop type classification, full fine-tuning shows consistent improvement and better generalization.

Table 2. Effect of loss functions on model performance under full fine-tuning strategy with 100% of the dataset. Metrics highlight Dice Loss's superior performance, particularly regarding macro F1 scores and IoU, for both tasks.

Task	Loss Function	F1 Score	IoU (Jaccard)	Accuracy
Crop Type Classification	Cross-Entropy	0.69	0.55	0.73
	Focal Loss	0.62	0.58	0.76
	Dice Loss	0.79	0.67	0.82
Crop Health Classification	Cross-Entropy	0.17	0.18	0.75
	Focal Loss	0.18	0.18	0.70
	Dice Loss	0.32	0.25	0.71

5. Discussion and Conclusion

This study highlights the effectiveness of fully fine-tuning the Prithvi-EO-2.0 encoder for agricultural applications such as crop type and crop health classification. Our ablation results show that fully fine-tuning the encoder significantly enhances crop type classification accuracy. This improvement is reflected in faster convergence and lower loss

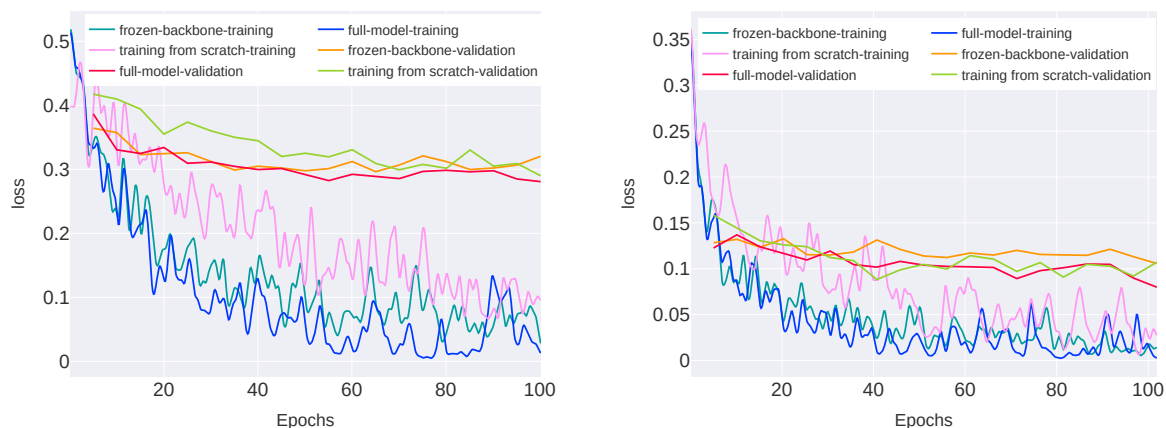


Fig. 5. Dice Loss curves during training for crop health (left) and crop type (right) classification tasks. Full fine-tuning consistently achieves lower training and validation losses compared to other strategies, indicating better optimization and convergence behavior.

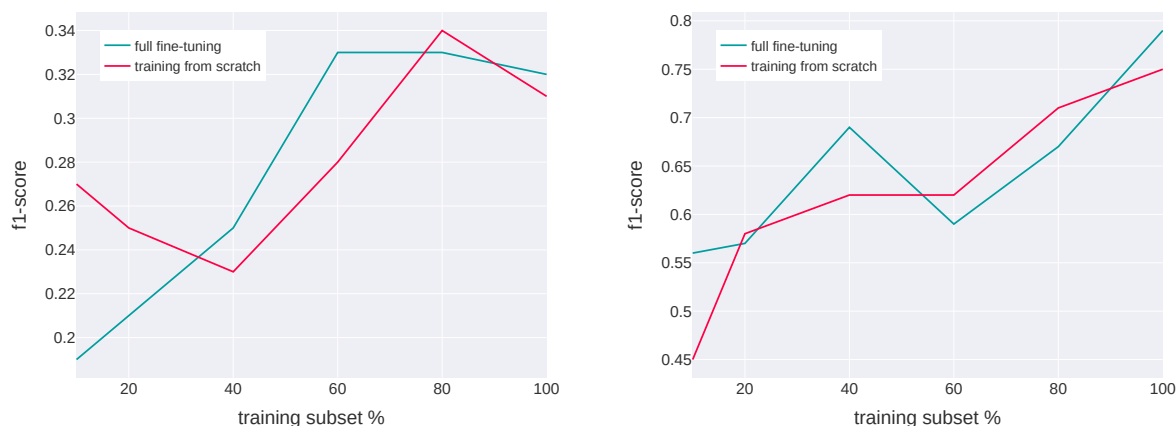


Fig. 6. Effect of dataset size on macro F1 scores for crop health (left) and crop type (right) classification tasks. Pretrained models with full fine-tuning exhibit significant performance advantages in lower-data scenarios (20%–40% dataset) for crop type classification and intermediate data availability (40%–60%) for crop health classification. Performance gains stagnate beyond these ranges for pretrained models, whereas models trained from scratch continue to improve steadily as more data becomes available.

values (Figures 4, 5), demonstrating the benefit of adapting all layers of the model to domain-specific data. However, for crop health classification, where class imbalance is severe, fine-tuning also introduces risks of overfitting. This indicates that while foundation models offer strong performance, their deployment in imbalanced settings requires careful calibration.

Interestingly, we observed that pretrained models occasionally underperformed compared to models trained from scratch, particularly in crop health scenarios. This unexpected behavior highlights the complexity of transfer learning in imbalanced agricultural contexts. It motivates applying explainable AI techniques such as Layerwise Relevance Propagation (LRP)[2] and Spectral Relevance Analysis [8] to gain deeper insights into model behavior. Such analyses will particularly help identify feature-level influences for correctly and incorrectly classified minority-class instances, thereby guiding targeted model improvements.

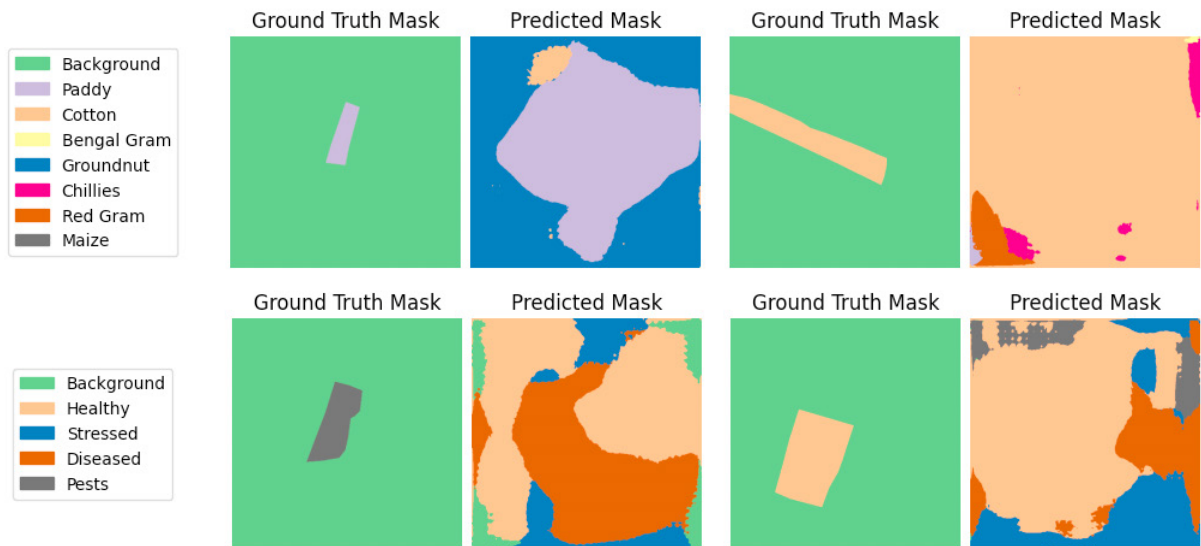


Fig. 7. Inference results demonstrating the model's capability in identifying and classifying crop characteristics: (Top row) Crop type classification, showing the model's predictions against ground truth labels across multiple crop varieties, emphasizing areas of accurate segmentation and regions requiring improvement; (Bottom row) Crop health classification, comparing ground truth and predicted masks for various health conditions including pest-affected (left) which was predicted as diseased, and correctly predicted healthy crops right.

Despite promising results, several limitations persist. A key challenge is the resolution mismatch between available remote sensing data and the small plot sizes typical of smallholder farms. While resampling 30-meter HLS imagery to 0.5 meters improves spatial alignment, it does not resolve the core issue of coarse resolution, which can impair model performance for small and fragmented fields. Future work will explore the use of higher-resolution Sentinel-2 data (10 meters) to improve spatial fidelity, even at the expense of reduced temporal coverage. This trade-off must be carefully assessed to ensure effective and timely crop monitoring.

Beyond technical performance, achieving real-world impact requires a focus on usability and accessibility. To that end, we plan to conduct a human-centered usability study involving farmers and agricultural stakeholders. Their feedback will inform the design of practical tools and help align AI solutions with the realities of farming communities. In addition, integrating ground-based observations will support ongoing model refinement and enhance operational relevance.

In summary, our work demonstrates the potential of fine-tuned foundation models for agricultural monitoring, while also identifying important limitations and future directions. Successfully addressing these challenges will be key to translating AI advances into field-ready tools that are inclusive, scalable, and genuinely beneficial to smallholder farmers.

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